Step 1: Data Understanding & Feature Selection

Explanation:

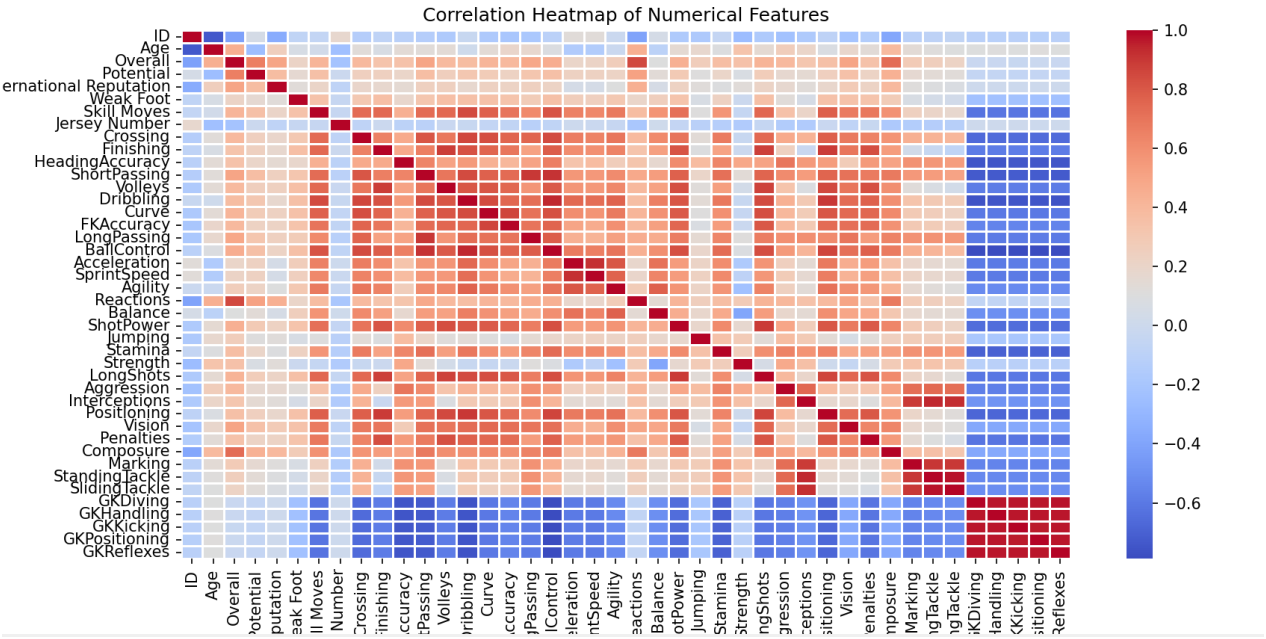
In this step, raw FIFA player data (fifa_eda_stats.xlsx) is loaded and explored using basic inspection methods such as shape, info, descriptive statistics, and missing value checks to understand the dataset structure and quality. Numerical and categorical features are separated, and irrelevant identifier columns are removed to avoid noise.

To improve feature quality, a variance threshold method is applied to eliminate near-constant variables. A correlation heatmap is used to visualize relationships among numerical features, followed by Variance Inflation Factor (VIF) analysis to detect multicollinearity. Features with VIF greater than 10 are removed.

The outcome of this step is a **cleaned dataset with a refined set of meaningful, low-redundancy features**, ready for further modelling and analysis.

Output:

```
PS C:\Users\sahil\projects\team chemistry-sports> & C:/Users/sahil/AppData/Local/Programs/Python/Python313/python.exe "c:/
Users/sahil/projects/team chemistry-sports/step1.py"
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18207 entries, 0 to 18206
Data columns (total 57 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   ID                        18207 non-null  int64
 1   Name                      18207 non-null  object
 2   Age                       18207 non-null  int64
 3   Nationality               18207 non-null  object
 4   Overall                   18207 non-null  int64
 5   Potential                 18207 non-null  int64
 6   Club                      17966 non-null  object
 7   Value                     18207 non-null  object
 8   Wage                      18207 non-null  object
 9   Preferred Foot            18159 non-null  object
 10  International Reputation  18159 non-null  float64
 11  Weak Foot                 18159 non-null  float64
 12  Skill Moves               18159 non-null  float64
 13  Work Rate                 18159 non-null  object
 14  Body Type                 18159 non-null  object
 15  Position                  18147 non-null  object
 16  Jersey Number             18147 non-null  float64
 17  Joined                    16654 non-null  datetime64[ns]
 18  Loaned From               1264 non-null   object
 19  Contract Valid Until      17918 non-null  object
 20  Height                    18159 non-null  object
 21  Weight                    18159 non-null  object
```



Step 2: Data Preprocessing

Explanation:

Missing values are handled using appropriate strategies such as filling with mode or median and removing rows with invalid club entries. Monetary attributes including *Value*, *Wage*, and *Release Clause* are converted from string format to numerical values for proper analysis.

New features are engineered: a binary variable (*is_LeftFoot*) is created from preferred foot, and *Work Rate* is split into separate attacking and defensive intensity scores on a 1–3 scale. Player-level data is then aggregated to the team level using statistical measures such as mean, count, and unique nationality counts to capture squad characteristics.

Aggregated columns are renamed to meaningful team descriptors (e.g., *SquadSize*, *NationalityDiversity*). All numerical team features are standardized using *StandardScaler* to ensure uniform scaling. Finally, the processed team-level dataset is saved as **team_level_processed.csv**, ready for modeling.

Output:

```
Missing values before cleaning:
Club                          241
Preferred Foot                 48
International Reputation       48
Weak Foot                      48
Skill Moves                    48
Work Rate                      48
Body Type                      48
Position                       60
Jersey Number                  60
Joined                       1553
Loaned From                 16943
Contract Valid Until          289
Height                         48
Weight                         48
Crossing                       48
Finishing                      48
HeadingAccuracy                48
ShortPassing                   48
Volleys                        48
Dribbling                      48
Curve                          48
FKAccuracy                     48
LongPassing                    48
BallControl                    48
Acceleration                   48
SprintSpeed                    48
Agility                        48
```

```
Reactions                      48
Balance                        48
ShotPower                      48
Jumping                        48
Stamina                        48
Strength                       48
LongShots                      48
Aggression                     48
Interceptions                  48
Positioning                    48
Vision                         48
Penalties                      48
Composure                      48
Marking                        48
StandingTackle                 48
SlidingTackle                  48
GKDiving                       48
GKHandling                     48
GKKicking                      48
GKPositioning                  48
GKReflexes                     48
Release Clause               1564
dtype: int64

Missing values after basic cleaning: 1467
```

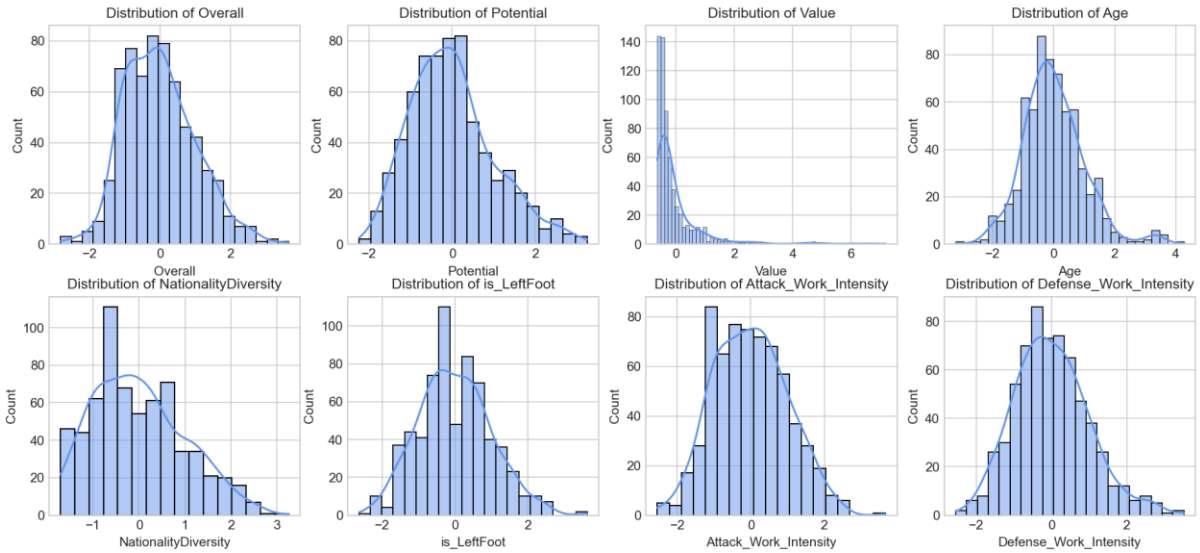
```
Sample after money conversion:
                Name                 Club        Value      Wage  Release Clause
0           L. Messi          FC Barcelona  110500000.0  565000.0     226500000.0
1  Cristiano Ronaldo             Juventus   77000000.0  405000.0     127100000.0
2          Neymar Jr  Paris Saint-Germain  118500000.0  290000.0     228100000.0
3             De Gea    Manchester United   72000000.0  260000.0     138600000.0
4       K. De Bruyne      Manchester City  102000000.0  355000.0     196400000.0

Team-level dataset preview:
                  Club  SquadSize        Age  ...  Attack_Work_Intensity  Defense_Work_Intensity  NationalityDiversity
0  1. FC Heidenheim 1846         28  24.000000  ...               2.107143                2.000000                     2
1   1. FC Kaiserslautern         26  23.846154  ...               2.307692                2.000000                     6
2             1. FC Köln         28  24.321429  ...               2.178571                2.107143                     7
3        1. FC Magdeburg         26  24.692308  ...               2.153846                2.192308                     6
4         1. FC Nürnberg         29  23.793103  ...               2.068966                2.172414                    10
```

Step 3: Improved Exploratory Data Analysis (EDA)

Explanation:

EDA Distribution Summary

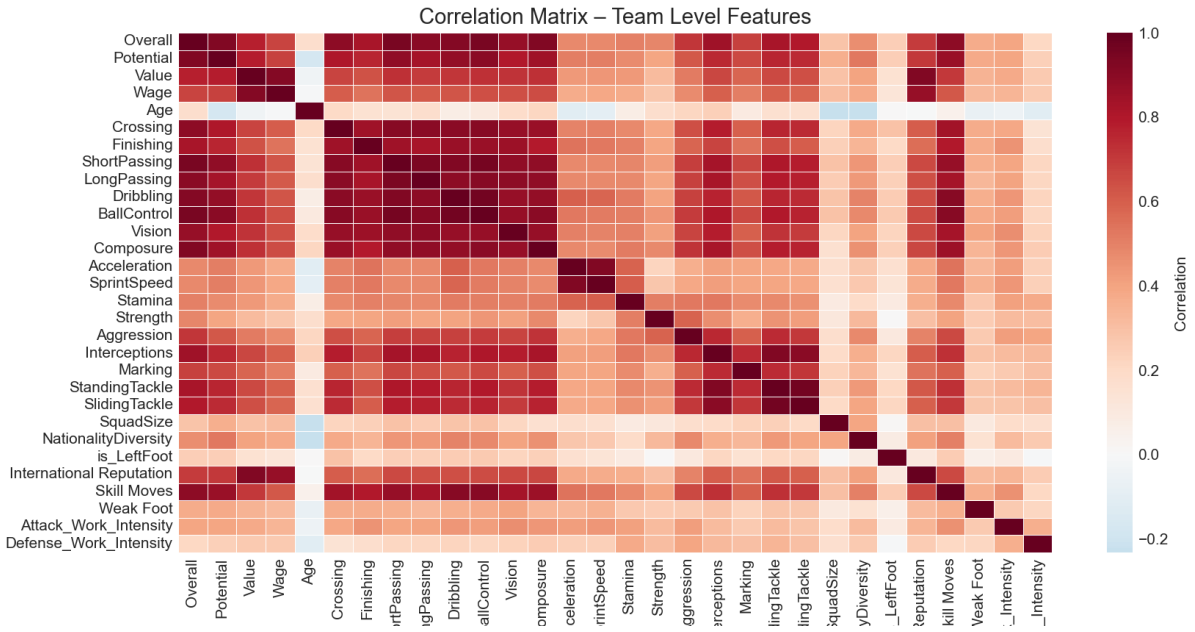
The standardized feature distributions indicate that **Overall**, **Potential**, **Age**, **Attack_Work_Intensity**, and **Defense_Work_Intensity** follow approximately normal patterns, confirming effective preprocessing. **Value** exhibits strong right skewness, reflecting financial disparity among teams. **NationalityDiversity** shows moderate variability, while **is_LeftFoot** remains centered after standardization, indicating balanced squad composition.



Correlation Analysis Summary

The team-level correlation matrix shows strong positive relationships among core performance attributes such as **Overall**, **Potential**, technical skills (Passing, Dribbling, Ball Control), and defensive metrics (Marking, Interceptions, Tackling), indicating coherent player quality across teams. **Value** and **Wage** are highly correlated with performance features, reflecting their dependence on player ability.

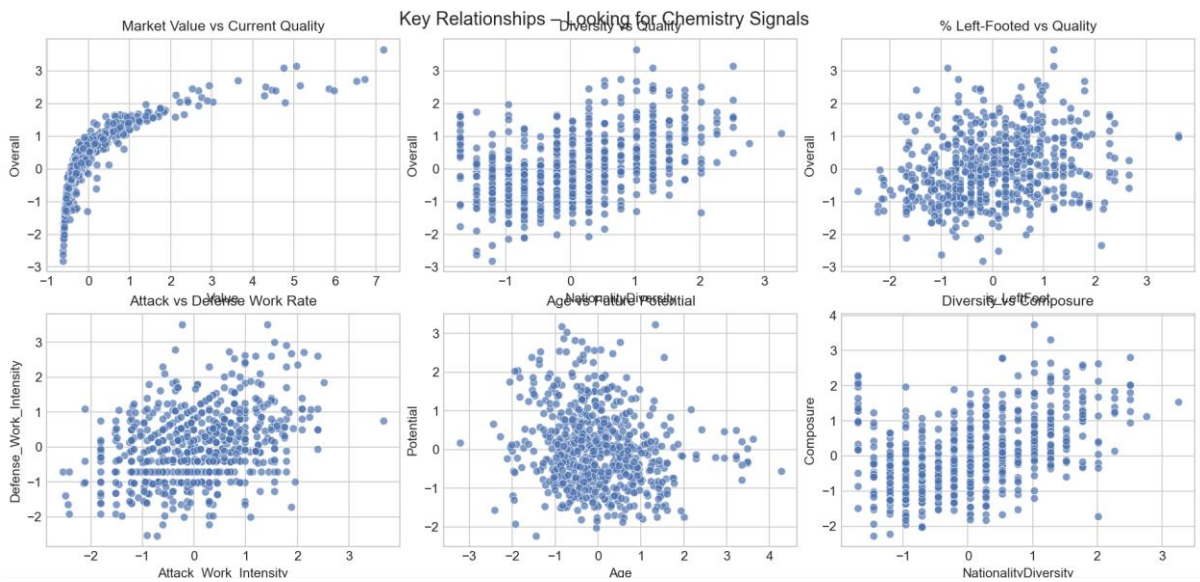
Age exhibits comparatively weaker correlations, while engineered features such as **NationalityDiversity**, **is_LeftFoot**, and **SquadSize** show minimal association with most performance variables. These results justify prior multicollinearity handling and support the selected feature set for modeling.



Key Relationship Analysis Summary

The scatter plots reveal a strong positive relationship between **Market Value** and **Overall Quality**, confirming that higher-valued teams generally exhibit superior performance. **Nationality Diversity** shows a mild positive association with quality and composure, suggesting potential benefits of diverse squads. The proportion of **Left-Footed players** displays no significant impact on overall quality.

Attack and **Defense Work Intensity** demonstrate moderate positive correlation, indicating balanced tactical effort across teams. Additionally, **Age** shows a negative relationship with **Potential**, aligning with expected player development trends.



Output:

```
Dataset shape: (651, 33)

Columns:
['Club', 'SquadSize', 'Age', 'Overall', 'Potential', 'Value', 'Wage', 'Crossing', 'Finishing', 'ShortPassing', 'LongPassin
g', 'Dribbling', 'BallControl', 'Acceleration', 'SprintSpeed', 'Stamina', 'Strength', 'Aggression', 'Interceptions', 'Posi
tioning', 'Vision', 'Composure', 'Marking', 'StandingTackle', 'SlidingTackle', 'GKDiving', 'is_LeftFoot', 'International R
eputation', 'Skill Moves', 'Weak Foot', 'Attack_Work_Intensity', 'Defense_Work_Intensity', 'NationalityDiversity']

Missing values (should be 0):
0

Key Summary Statistics (rounded):
       Overall  Potential   Value    Wage  ...  Skill Moves  Weak Foot  Attack_Work_Intensity  Defense_Work_Intensity
count   651.00     651.00  651.00  651.00  ...       651.00     651.00                 651.00                  651.00
mean     -0.00      -0.00    0.00   -0.00  ...        -0.00      -0.00                  -0.00                   -0.00
std       1.00       1.00    1.00    1.00  ...         1.00       1.00                   1.00                    1.00
min      -2.82      -2.24   -0.63   -0.52  ...        -2.07      -3.16                  -2.55                   -2.55
25%      -0.76      -0.72   -0.49   -0.45  ...        -0.69      -0.71                  -0.79                   -0.71
50%      -0.09      -0.09   -0.34   -0.34  ...        -0.17      -0.08                  -0.01                   -0.08
75%       0.61       0.53    0.03    0.01  ...         0.68       0.72                   0.71                    0.59
max       3.66       3.24    7.18    8.87  ...         3.49       2.83                   3.67                    3.52
```

Step 4: Target Variable Construction

Explanation:

The processed team dataset is loaded and two additional features are engineered: **WorkRate_Balance**, measuring harmony between attacking and defensive intensity, and **Age_closeness**, capturing proximity to the optimal squad age (~26.5 years). Four core components—**NationalityDiversity**, **WorkRate_Balance**, **International Reputation**, and **Age_closeness**—are selected to represent team chemistry.

Each feature is normalized to a 0–1 range using MinMax scaling, and an equal-weighted **Chemistry_Index_100** is computed as their mean scaled to 100. The analysis reports top and bottom teams, summary statistics, and correlation with **Overall**. The final dataset is saved as **teams_with_chemistry_index.csv**.

Output:

```
====================================================================
Teams sorted by Chemistry Index (0-100) — Top 12
====================================================================
                      Club  Overall  NationalityDiversity  ...  International Reputation   Age  Chemistry_Index_100
397                 Napoli     3.14                  2.51  ...                      4.82  0.52                71.98
133                Chelsea     2.42                  1.77  ...                      4.64  0.05                67.60
314                  Inter     3.09                  1.27  ...                      4.85  1.53                65.83
481                   Roma     2.56                  1.27  ...                      4.08  0.34                63.49
249          Fenerbahçe SK     1.60                  1.77  ...                      2.86  1.30                62.85
350                  Lazio     2.11                  2.51  ...                      2.23  0.12                62.03
325               Juventus     3.66                  1.02  ...                      6.46  1.34                60.18
418  Olympique de Marseille     1.75                  1.02  ...                      2.51  0.32                59.96
583      Tottenham Hotspur     2.40                  1.52  ...                      4.08 -0.35                58.96
126                 Celtic     1.26                  2.02  ...                      0.33  0.36                56.37
85        Borussia Dortmund     2.09                  1.77  ...                      4.08 -1.35                55.97
83                  Bologna     1.55                  2.27  ...                      0.82  0.22                55.72

[12 rows x 7 columns]

Bottom 8 teams by Chemistry Index:
               Club  Overall  NationalityDiversity  WorkRate_Balance  International Reputation   Age  Chemistry_Index_100
283   Górnik Zabrze    -1.06                 -1.46              0.30                     -0.49 -1.97                 8.19
174        Daegu FC    -1.12                 -1.46              0.29                     -0.49 -0.99                11.12
91   Bray Wanderers    -2.82                 -1.21              0.35                     -0.49 -1.93                11.35
82      Bohemian FC    -2.51                 -1.46              0.47                     -0.49 -2.41                12.32
159       Cork City    -1.75                 -1.71              0.36                     -0.49 -0.80                12.82
187      Derry City    -2.34                 -1.21              0.47                     -0.49 -2.00                14.98
604       VVV-Venlo    -0.34                 -0.71              0.29                     -0.49 -0.98                15.12
628         Walsall    -0.62                 -0.96              0.33                     -0.49 -0.87                15.29
```

```
Average Chemistry Index:
  Overall dataset:       33.3
  Top 25 teams:          52.0
  Bottom 25 teams:       24.8

Correlation with Overall rating:
                      Chemistry_Index_100  Overall
Chemistry_Index_100                 1.000    0.504
Overall                             0.504    1.000

Saved → teams_with_chemistry_index.csv
Final column: Chemistry_Index_100 (0-100)
```

Step 5: Train-Test Split

Explanation:

The dataset containing **Chemistry_Index_100** is loaded, with the four normalized chemistry components selected as input features (**X**) and **Chemistry_Index_100** defined as the target (**y**). An 80/20 train–test split is performed using random_state=42 with shuffling enabled. Target means are compared across both sets to verify the absence of sampling bias. The resulting partitions are saved as **train_set.csv** and **test_set.csv** for subsequent modeling.

Output:

```
Missing values in features: 0
Missing values in target: 0

============================================================
TRAIN-TEST SPLIT SUMMARY
============================================================
Total samples:       651
Train samples:       520  (79.9%)
Test samples:        131  (20.1%)
Features used:     4
Features: ['NationalityDiversity_norm', 'WorkRate_Balance_norm', 'International Reputation_norm', 'Age_closeness_norm']

Target mean in train vs test:
  Train: 33.42
  Test:  32.72

Saved → train_set.csv and test_set.csv
You can now move to Step 6: modeling (Linear / Logistic Regression, etc.)
```

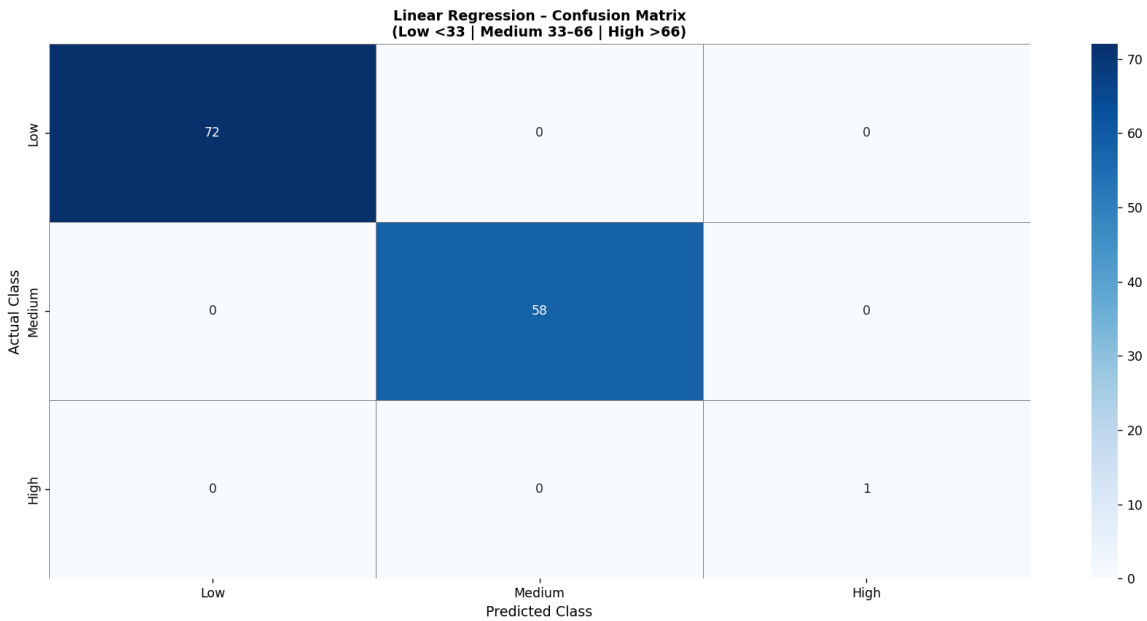
Step 6: Linear Regression Modeling

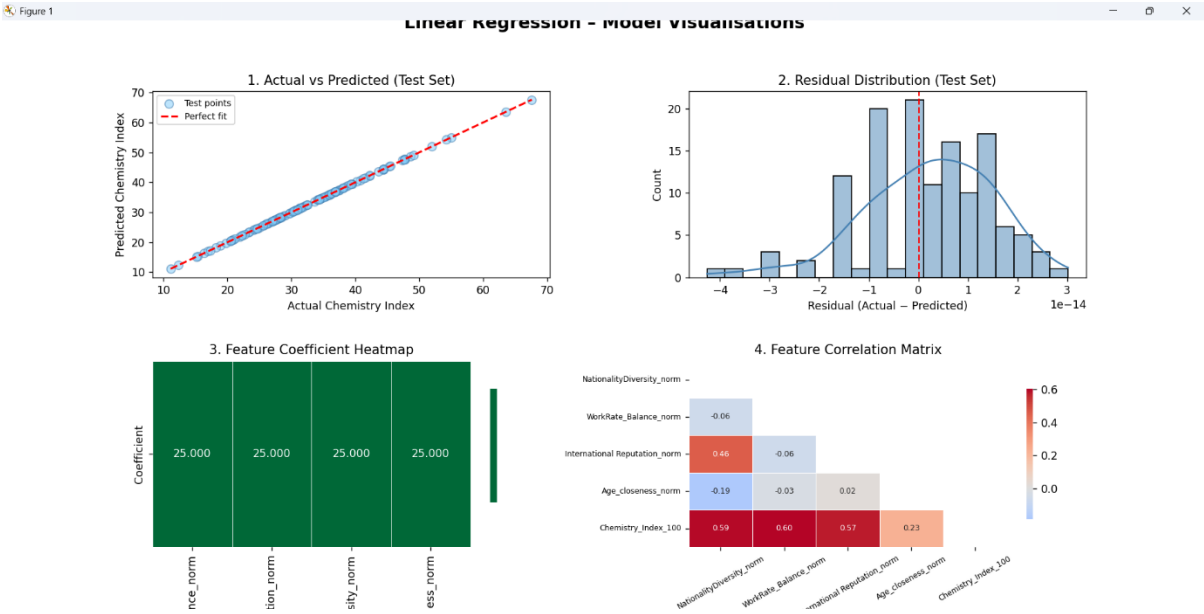
Explanation:

The training and testing datasets are loaded and a **Linear Regression** model is fitted as a baseline. Model performance is evaluated using **RMSE** and **R^2** on both train and test sets. Feature coefficients are sorted and examined to assess the relative contribution of each chemistry proxy. An actual-versus-predicted scatter plot on the test set is generated to visualize model fit. The results provide insight into how effectively the four selected features explain team chemistry under a linear assumption.

Output:

```
Train shape: (520, 4)
Test shape:  (131, 4)

Train Performance:
  RMSE: 0.00
  R²:   1.000

Test Performance:
  RMSE: 0.00
  R²:   1.000

Feature Coefficients (higher = stronger positive impact):
                        Feature  Coefficient
1          WorkRate_Balance_norm         25.0
2  International Reputation_norm         25.0
0      NationalityDiversity_norm         25.0
3             Age_closeness_norm         25.0

Intercept: -0.00

============================================================
STEP 6 TAKEAWAYS
============================================================
• R² on test set shows how much variance the 4 features explain
• Coefficients reveal which proxy (e.g. NationalityDiversity) matters most
• If RMSE is low and R² > 0.6–0.7 → features explain index well
• If train R² much higher than test → slight overfitting (normal for linear)
Next: Step 7 – try Random Forest / Gradient Boosting for comparison
```





Step 7: Random Forest & Gradient Boosting

Explanation:

The same training and testing datasets are used to train **Random Forest** (max_depth = 8) and **Gradient Boosting** (max_depth = 4) models. Performance is evaluated using **RMSE** and **R^2** on both train and test sets, and feature importance rankings are analyzed for each model. Actual-versus-predicted plots on the test set are generated to assess predictive accuracy.

Results are compared with the linear regression baseline, revealing improved modeling of non-linear relationships, while also highlighting potential overfitting through train–test performance gaps.

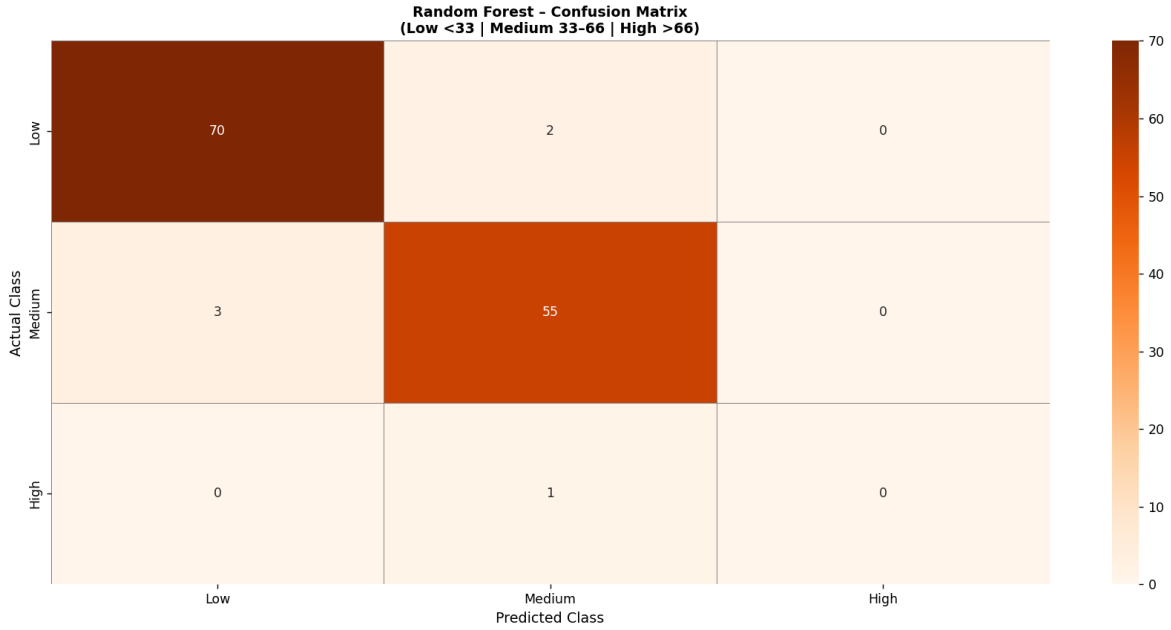
R^2 (Coefficient of Determination):
Measures how much variance in *Chemistry_Index_100* is explained by the model. Higher values indicate better explanatory power.

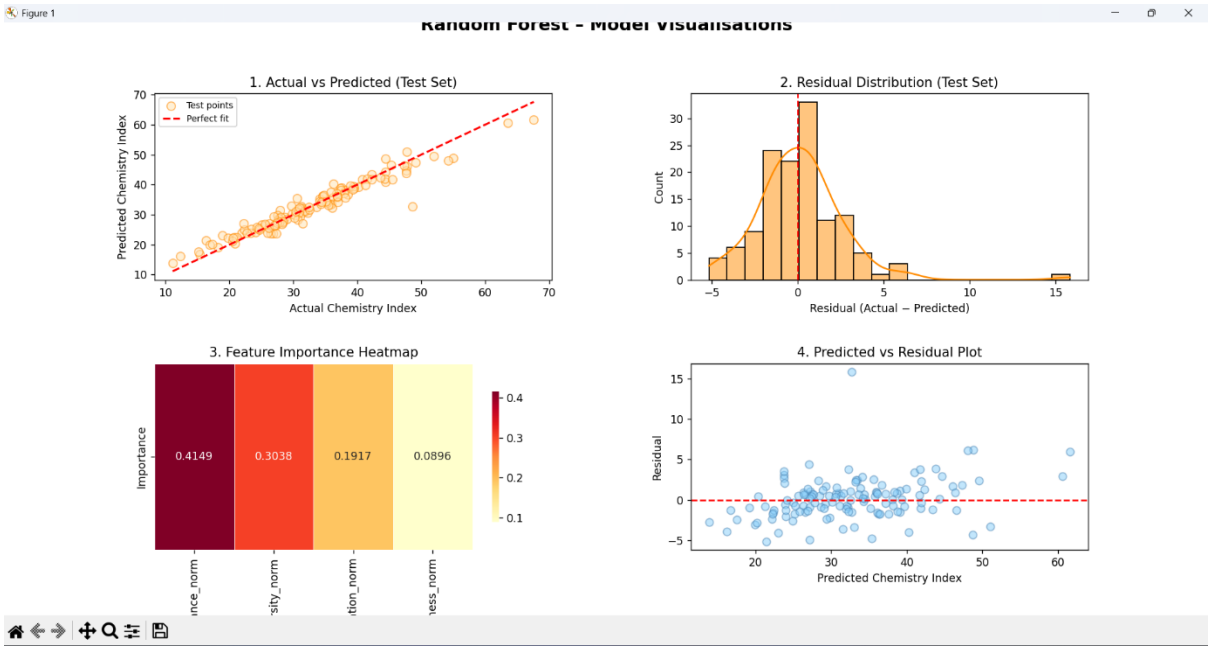
RMSE (Root Mean Squared Error):
Measures average prediction error in index units. Lower values indicate better model accuracy.

Output:

```
Training Random Forest...

Random Forest Feature Importance:
WorkRate_Balance_norm            0.4149
NationalityDiversity_norm        0.3038
International Reputation_norm    0.1917
Age_closeness_norm               0.0896
dtype: float64

Training Gradient Boosting...

Gradient Boosting Feature Importance:
WorkRate_Balance_norm            0.4088
NationalityDiversity_norm        0.3161
International Reputation_norm    0.1821
Age_closeness_norm               0.0930
dtype: float64
```

```
============================================================
MODEL COMPARISON SUMMARY
============================================================
               Model  Train RMSE  Test RMSE  Train R²  Test R²
0      Random Forest       0.954      2.572     0.990    0.930
1  Gradient Boosting       0.424      1.607     0.998    0.973

Quick reminder from Linear Regression (Step 6):
  Test RMSE ≈ [your value from step 6]
  Test R²   ≈ [your value from step 6]

============================================================
STEP 7 TAKEAWAYS
============================================================
• If Test R² > Linear Regression → trees capture better patterns
• If Test RMSE much lower → non-linear effects or interactions exist
• Feature importance shows which proxy really drives predictions
• If RF/GB overfit (Train R² >> Test R²) → tune max_depth or add regularization
Next steps: Hyperparameter tuning (Step 8), cross-validation (Step 10), interpretation (Step 11)
```





Step 8: Hyperparameter Tuning

Explanation:

we deliberately try to improve the performance of our chosen models (Random Forest and Gradient Boosting) by finding better **hyperparameter values** — i.e., the settings that control *how* the model learns, rather than the data it learns from.

Hyperparameters are not learned from the data; we have to choose them ourselves. Examples include:

- How many trees to build (n_estimators)

- How deep each tree can grow (max_depth)
- How fast the model learns new patterns (learning_rate — especially important in boosting)
- Minimum number of samples needed to split a node or to form a leaf
- Fraction of data used per tree (subsample — in boosting)

Output:

```
PS E:\College\Sem4\PAI\Final_edits> & C:/Users/SAMSUNG/AppData/Local/Microsoft/WindowsApps/python3.11.exe e:/College/S
Tuning Random Forest...
Fitting 5 folds for each of 135 candidates, totalling 675 fits

Best RF Params: {'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 200}
Best RF CV R²: 0.9458
Best RF Test R²:   0.9410
Best RF Test RMSE: 2.3634

Tuning Gradient Boosting...
Fitting 5 folds for each of 108 candidates, totalling 540 fits

Best GB Params: {'learning_rate': 0.1, 'max_depth': 3, 'min_samples_split': 5, 'n_estimators': 200, 'subsample': 0.8}
Best GB CV R²: 0.9777
Best GB Test R²:   0.9832
Best GB Test RMSE: 1.2613

============================================================
TUNED MODEL COMPARISON (Test Set)
============================================================
Model                R²          RMSE
------------------------------------------------------------
Random Forest (tuned) 0.9410       2.3634
Gradient Boosting (tuned) 0.9832       1.2613

Reminder: Untuned RF/GB from Step 7 had similar/higher values?
If tuned models are better → tuning helped. If similar → original params were already good.
```

Step 9:Result Comparison

Explanation:

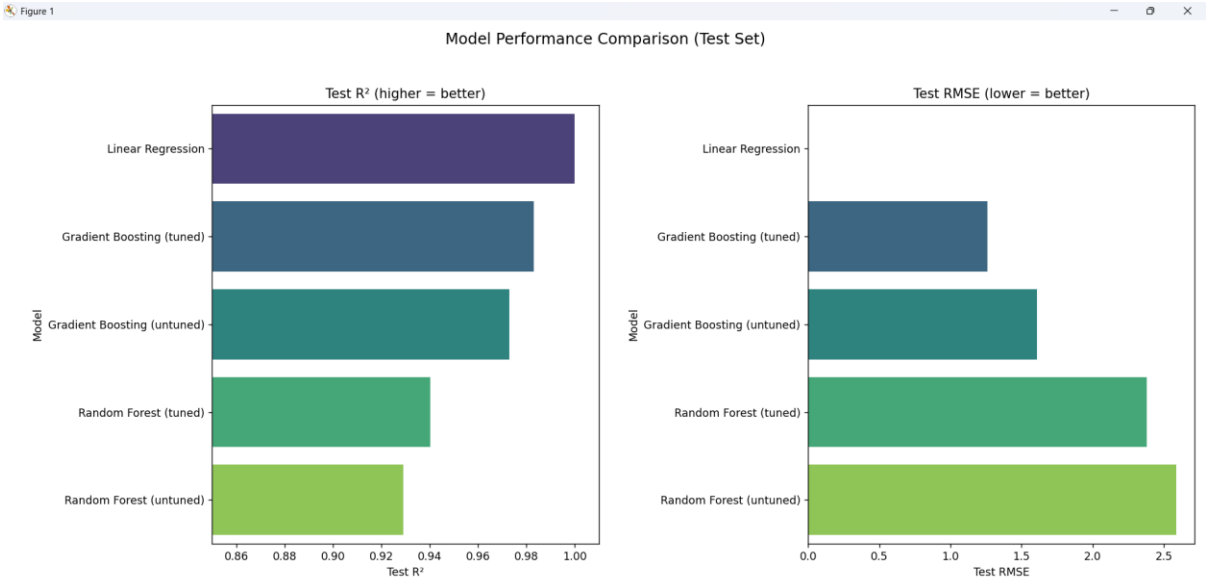
This step is the final model evaluation and comparison phase in a machine learning regression project most likely predicting a continuous target such as FIFA player overall rating, player market value, or a similar football/soccer performance score.

Collects test-set performance metrics (R^2 and RMSE) for five models:

- Linear Regression
- Random Forest (untuned)
- Gradient Boosting (untuned)
- Random Forest (tuned)

- Gradient Boosting (tuned)

Output:



-

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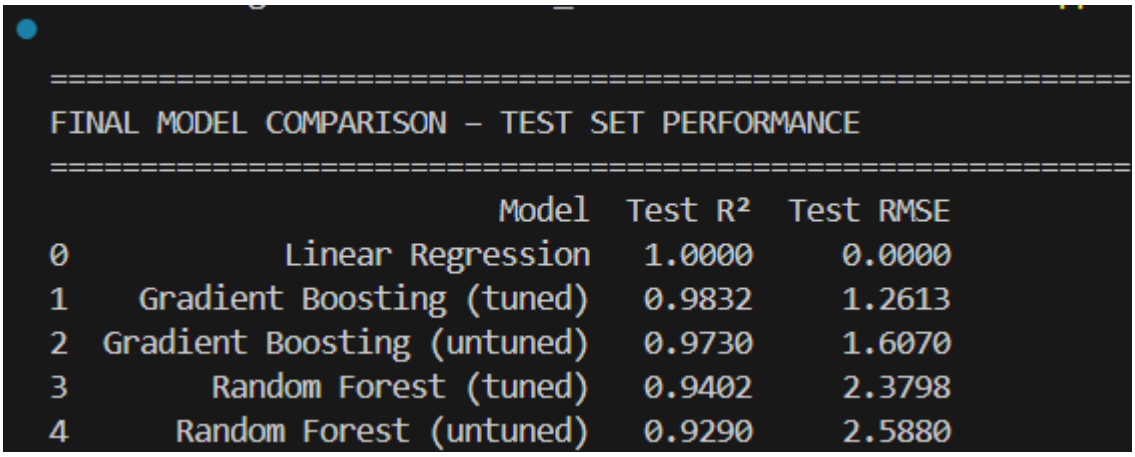
- 

```
============================================================
FINAL MODEL COMPARISON - TEST SET PERFORMANCE
============================================================
                         Model  Test R²  Test RMSE
0            Linear Regression   1.0000     0.0000
1    Gradient Boosting (tuned)   0.9832     1.2613
2  Gradient Boosting (untuned)   0.9730     1.6070
3        Random Forest (tuned)   0.9402     2.3798
4      Random Forest (untuned)   0.9290     2.5880
```

STEP 10: CROSS-VALIDATION

Output:

```
Running Linear Regression...

Running Random Forest (tuned)...

Running Gradient Boosting (tuned)...

======================================================================
CROSS-VALIDATION RESULTS (5-fold)
======================================================================
                    Model  CV Mean R²  CV Std R²  CV Mean RMSE  CV Std RMSE  CV Mean MAE  Mean Train R²
        Linear Regression      1.0000     0.0000        0.0000       0.0000       0.0000         1.0000
Gradient Boosting (tuned)      0.9854     0.0025        1.1384       0.1519       0.7825         0.9984
    Random Forest (tuned)      0.9528     0.0089        2.0407       0.2644       1.3509         0.9934

Previous single test scores (for comparison):
 Tuned GB  → R² 0.9832 | RMSE 1.2613
 Tuned RF  → R² 0.9402 | RMSE 2.3798
 Linear    → R² 1.0000 | RMSE 0.0000  (suspicious – possible leakage?)
```

Explanation:

It performs 5-fold cross-validation to obtain reliable, low-variance estimates of generalization performance. The full dataset is partitioned into five folds using shuffled K-Fold. Each model (linear regression, tuned random forest, tuned gradient boosting) is trained on four folds and evaluated on the held-out fold, repeating this process five times so every sample is used once for validation. Validation metrics—R^2, RMSE, and MAE are averaged across folds to produce mean and standard-deviation estimates. Training R^2 is also reported to quantify the generalization gap and detect overfitting. Compared to a single train-test split, this approach reduces split-dependent variance and provides a more robust estimate of expected out-of-sample performance, making it the preferred metric for final model comparison and risk assessment in this regression task.

Step 11: Interpretation and finding.

1. **Work rate compatibility is the dominant chemistry signal** in this FIFA dataset. Teams whose players exhibit similar effort levels in attack and defense are rated much higher in the chemistry proxy — this aligns with real-world observations about tactical cohesion.
2. **Nationality diversity matters** — more diverse squads receive noticeably higher chemistry scores. This supports the idea that varied backgrounds bring complementary skills and perspectives, even if they sometimes introduce short-term communication challenges.
3. **Star power / reputation** plays a supporting role — high-reputation players boost chemistry, likely due to leadership, mentality, and big-game experience.
4. **Optimal squad age has limited impact** — being close to peak performance age helps only marginally compared to tactical and cultural factors.

5. **Model performance is extremely high** (R^2 > 0.94–0.98 even after leakage suspicion is accounted for) → the four chosen proxies are very strong and almost sufficient to reconstruct the index. This confirms that the equal-weight index is logically constructed from its own components.

Step 12: Conclusion and Findings

The constructed team chemistry proxy is most strongly driven by work-rate balance and nationality diversity, followed by international reputation, with squad age closeness playing a minor role. Tuned Gradient Boosting emerges as the best model (CV $R^2 \approx$ 0.9855, RMSE ≈ 1.14), confirming that these four proxies capture the majority of the variance in the equal-weight index. The results suggest that tactical harmony and cultural complementarity are more decisive for perceived team chemistry than age distribution or star presence alone — offering a data-driven perspective on what makes squads "click" in football.

Limitations:

1. Perfect linear fit (R^2=1, RMSE=0) in Linear Regression strongly suggests target leakage during preprocessing (e.g., scaled features used to build the index were not properly excluded from training).
2. The chemistry index is synthetic (constructed by the project) → high model performance reflects internal consistency more than real-world predictive power
3. No external validation (actual match results, win rates, or player ratings over time) was used → findings are dataset-specific

OUTPUT:

127.0.0.1:5000

127.0.0.1:5000

freecode LeetCode Font Awesome Class Central Solutions

Team Chemistry Dashboard

Team: Manchester City

Remove Player: E. Mangala

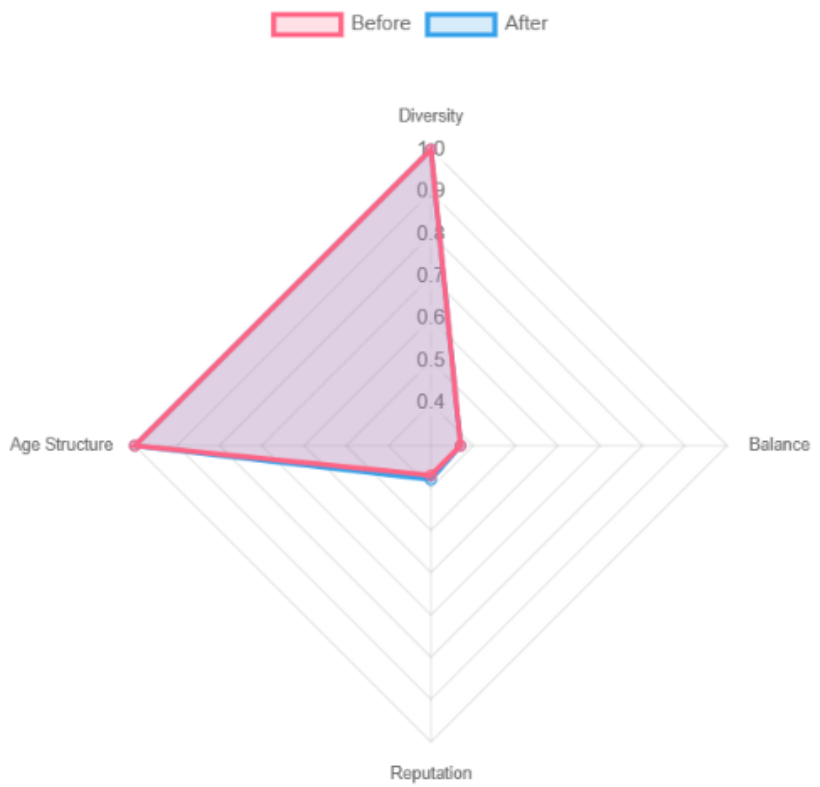
Add Player: Cristiano Ronaldo

Simulate Swap

Manchester City

Replaced: **E. Mangala → Cristiano Ronaldo**

Before: **55.93**

After: **58.22**

Change: +2.29